Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 5 — Exponentials and logarithms Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第5章 · 指数与对数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C05-R01 · P3 · Chapter 5

For $x > 1$, $\ln(x+1) - \ln(x-1) = \ln 3$.

(a) Find x exactly. (3)

(b) Find the exact value of $\ln(x^2 - 1)$ at this solution. (2)

(Total for Question 1 is 5 marks)

A cooling model is $T = 18 + 62e^{-0.2t}$, where $t \geq 0$ is in minutes and T is in °C.

(a) Find the initial temperature and limiting temperature. (2)

(b) Find the time when $T = 30$ °C, giving an exact expression and a value to 2 decimal places. (4)

(Total for Question 2 is 6 marks)

3.

$e^{2x} - 5e^{x} + 4 = 0.$

(a) Find all exact solutions. (4)

(b) Find all real solutions of $e^{2x} - 5e^{x} + 4 < 0$. (3)

(Total for Question 3 is 7 marks)

Two quantities are related by $y = Ax^n$, with $A > 0, x > 0$. A plot of $\ln y$ against $\ln x$ is a straight line through $(0, \ln 3)$ and $(\ln 2, \ln 24)$.

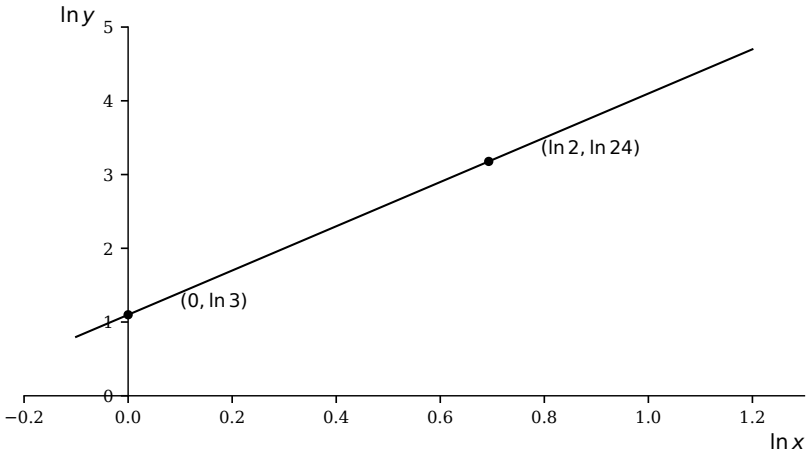


Figure 4

(a) Find A, n. (4)

(b) Find x when $y = 81$. (2)

(Total for Question 4 is 6 marks)

5.

A quantity satisfies $M = Ae^{kt}$. It has values 12 at $t = 1$ and 96 at $t = 4$.

(a) Find exact A, k. (4)

(b) Find the time when $M = 150$, to 3 decimal places. (3)

(Total for Question 5 is 7 marks)

6. P3-C05-R06 · P3 · Chapter 5

The curve is $y = \ln(4 - x)$.

(a) State its domain, vertical asymptote and axis intersections. (3)

(b) Find all x for which $y > \ln 2$. (3)

(Total for Question 6 is 6 marks)

$\ln x + \ln(x - 2) = \ln 8$, where $x > 2$.

(a) Solve exactly. (4)

(b) Find the exact value of $\ln x - \ln(x - 2)$ at the solution. (2)

(Total for Question 7 is 6 marks)

8. P3-C05-R08 · P3 · Chapter 5

A material has mass $m = 60e^{-kt}$, where $k > 0$. After 8 days its mass is 45.

(a) Find k exactly. (3)

(b) Find its half-life and the time to reach one quarter of its initial mass, in exact form. (3)

(Total for Question 8 is 6 marks)

9. P3-C05-R09 · P3 · Chapter 5

For $x > 0$, $y = Be^{ax}$. A graph of $\ln y$ against x has gradient $-3/2$ and vertical intercept 2.

(a) Find a, B. (3)

(b) Find the exact positive x at which $y = 1$. (2)

(Total for Question 9 is 5 marks)

$2e^{2x} - 7e^{x} - 4 = 0$.

(a) Find all exact real solutions, showing why one substituted root is inadmissible. (4)

(b) Find the value of e^{-x} at the solution. (2)

(Total for Question 10 is 6 marks)

The function is $f(x) = \ln(x^2 + 1)$.

(a) State its domain, range and every axis intersection. (4)

(b) Solve $f(x) = \ln 5$ and $f(x) < \ln 5$ separately. (3)

(Total for Question 11 is 7 marks)

A growth model is $P = 200e^{kt}$, with t in years. The population doubles every 3 years.

(a) Find k exactly. (2)

(b) For this model, find

(i) the population after 9 years, (2)

(ii) the least whole number of years at which the population is at least 1000. (3)

(Total for Question 12 is 7 marks)

$\ln(x+3) = 2\ln x$, where $x > 0$.

(a) Find the exact solution. (4)

(b) Find the exact value of $\ln((x+3)/x)$ at the solution. (2)

(Total for Question 13 is 6 marks)

The curve is $y = Ae^{2x} + B$. Its horizontal asymptote is $y = -3$, and it passes through $(\ln 2, 13)$.

(a) Find A, B. (3)

(b) Find its exact x-intercept and y-intercept. (3)

(Total for Question 14 is 6 marks)

For $x > 1$, $F(x) = \ln\left(\dfrac{x+1}{x-1}\right)$.

(a) Show that $F(x) > 0$. (3)

(b) Find the input at which $F(x) = \ln 5$. (3)

(Total for Question 15 is 6 marks)

A signal follows $S = 12e^{-0.4t}$ and a background signal follows $B = 3e^{-0.1t}$, for $t \geq 0$.

(a) Find exactly when $S = B$. (3)

(b) Find the interval of time when $S > B$. (3)

(Total for Question 16 is 6 marks)

A relation $y = Ae^{kx}$ has positive observations $(0, 5)$ and $(2, 20)$.

(a) Find A, k exactly. (3)

(b) Find $\ln y$ at $x = -1$ and give the corresponding exact y. (3)

(Total for Question 17 is 6 marks)

$\ln x = \ln(2 - x) + \ln 3$.

(a) State the domain restrictions and solve. (4)

(b) Find $\ln[x(2 - x)]$ at the solution. (2)

(Total for Question 18 is 6 marks)

The resistance of a component is $R = R_0 e^{kT}$. It is 10 ohms at $T = 0$ and 15 ohms at $T = 20$.

(a) Find R_0, k exactly. (4)

(b) Find the temperature change that multiplies resistance by $3/2$, and explain why it is independent of the starting temperature. (3)

(Total for Question 19 is 7 marks)

The graph of $Y = \ln(y - 2)$ against x is a straight line with gradient $-1/2$ and intercept $\ln 6$.

(a) Find y in terms of x. (3)

(b) Find the exact x at which $y = 5$ and state the horizontal asymptote of the original graph. (3)

(Total for Question 20 is 6 marks)

$e^x + e^{-x} = 5/2$.

(a) Find all exact solutions. (4)

(b) Find all possible values of $e^x - e^{-x}$ at these solutions. (3)

(Total for Question 21 is 7 marks)

A quantity is modelled by $Q = 100(1 - e^{-kt})$, where $k > 0$, $t \geq 0$. At $t = 4$, $Q = 75$.

(a) Find k exactly and the limiting value of Q. (4)

(b) Find the time at which $Q = 93.75$. (3)

(Total for Question 22 is 7 marks)

Let $f(x) = \ln(3x - 2)$.

(a) Find f^{-1} with its domain and range. (4)

(b) Solve $f(x) = 2f(1)$. (3)

(Total for Question 23 is 7 marks)

For $x > 0$, $(\ln x)^2 - \ln x - 6 = 0$.

(a) Find all exact solutions. (4)

(b) Find all x for which $(\ln x)^2 - \ln x - 6 \leq 0$. (3)

(Total for Question 24 is 7 marks)

The curves $y = e^{2x}$ and $y = 9e^{-x}$ meet.

(a) Find the intersection exactly. (3)

(b) Find the horizontal distance between the intersection and the point on $y = e^{2x}$ where $y = 1$. (3)

(Total for Question 25 is 6 marks)

A power-law model is $y = Ax^n$. A base-10 log plot has intercept 2 and gradient -3.

(a) Find A, n and the value of y at $x = 2$. (4)

(b) State the gradient and intercept if natural logarithms are used instead on both axes. (2)

(Total for Question 26 is 6 marks)

27.

For real x, $F(x) = \ln(e^x + 1) - x$.

(a) Show that $F(x) = \ln(1 + e^{-x})$. (3)

(b) Find x when $F(x) = \ln 3$. (3)

(Total for Question 27 is 6 marks)

28. P3-C05-R28 · P3 · Chapter 5

A model for concentration is $C = 8e^{-0.3t} + 2$, with $t \geq 0$.

(a) Find the concentration after 5 time units, to 3 significant figures. (2)

(b) Find the set of times for which the concentration is below 3. Give the exact boundary time and its value to 2 decimal places. (4)

(Total for Question 28 is 6 marks)

The mass M grams of a sample at time t hours is modelled by $M = Ae^{-kt}$, where $A, k > 0$. The mass is 60 g at $t = 2$ and 24 g at $t = 5$.

(a) Find k and A, giving each to 4 significant figures. (3)

(b) Using unrounded constants,

(i) find the time at which the sample mass is 10 g, to 3 significant figures. (2)

(ii) find the length of time needed for any mass in this model to fall to one half of its value, and explain why the result does not depend on that mass. (3)

(Total for Question 29 is 8 marks)

Variables x, y satisfy $x > 0$, $y > 2$ and $y = 2 + px^q$, where $p, q > 0$. A graph of $\ln(y - 2)$ against $\ln x$ passes through $(0, \ln 3)$ and $(\ln 4, \ln 24)$.

(a) Find p and q exactly. (3)

(b) For this model,

(i) find x when $y = 83$. (2)

(ii) find the percentage increase in y when x increases from 1 to 1.21. Give your answer to 3 significant figures. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C05-R01 · 5 marks

(a) · 3 marks

$x = 2$.

合并商：(x+1)/(x−1)=3，解一次式，检查x>1。

M1	对数商法则
M1	线性求解
A1	合法根

(b) · 2 marks

$\ln 3$.

代x2得$x^2-1=3$，真数为正。

M1	正确代入
A1	值

Question 2

P3-C05-R02 · 6 marks

(a) · 2 marks

80 °C and 18 °C.

t0时指数1；t增大时衰减项趋0。

B1	初温
B1	极限温度

(b) · 4 marks

$t = 5\ln(31/6) \approx 8.21$ min.

62e^{−.2t}=12，取ln得−.2t=ln(6/31)。

M1	隔离指数
M1	自然对数
A1	精确时间
A1	精度

Question 3

P3-C05-R03 · 7 marks

(a) · 4 marks

$x = 0, \ln 4.$

u=e^x>0，(u−1)(u−4)=0，取ln回解。

M1	正变量代换
M1	因式分解
M1	还原
A1	两根

(b) · 3 marks

$0 < x < \ln 4.$

二次式在1

M1	二次不等式
M1	单调映回x
A1	区间

Question 4

P3-C05-R04 · 6 marks

(a) · 4 marks

$A = 3, n = 3.$

lny=nlnx+lnA，截距ln3；斜率ln(24/3)/ln2=3。

M1	正确线性化
M1	截距求A
M1	斜率求n
A1	参数

(b) · 2 marks

$x = 3.$

$81=3x^3$，x正给3。

M1	代回原模型
A1	x

Question 5

P3-C05-R05 · 7 marks

(a) · 4 marks

$k = \ln 2, A = 6.$

相除e^{3k}=8给kln2，再由Ae^k12求A6。

M1	两条件相除
M1	对数求k
M1	回求A
A1	精确参数

(b) · 3 marks

$t = \ln 25 / \ln 2 \approx 4.644.$

6e^{tln2}=150，tln2=ln25。

M1	正确阈值方程
M1	对数
A1	时间

Question 6

P3-C05-R06 · 6 marks

(a) · 3 marks

$x < 4$; $x = 4$; $(3, 0)$ and $(0, \ln 4)$.

真数4−x正；y0给4−x1；x0给ln4。

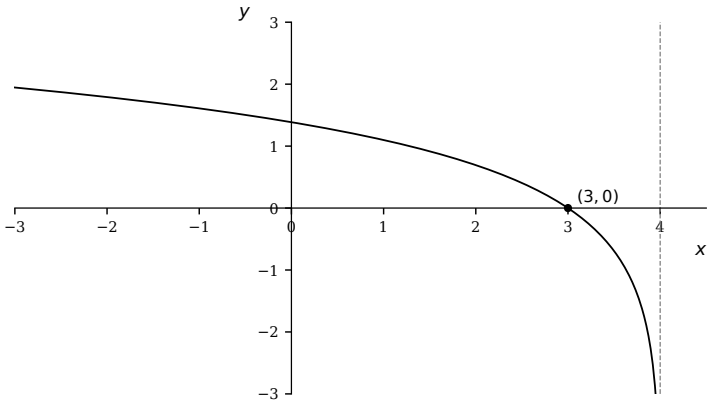
B1	域与渐近线
M1	横截距
A1	两点

(b) · 3 marks

$x < 2$.

ln严格递增，4−x>2，结果已包含在原域中。

M1	合法对数比较
M1	原域检查
A1	范围



Suggested sketch

Question 7

P3-C05-R07 · 6 marks

(a) · 4 marks

$x = 4$.

x(x−2)=8，方程(x−4)(x+2)=0，只留4。

M1	合并对数转乘积
M1	二次求根
M1	原域筛选
A1	x

(b) · 2 marks

$\ln 2$.

ln[4/2]=ln2，不能相减真数后直接取ln。

M1	对数商法则
A1	值

Question 8

P3-C05-R08 · 6 marks

(a) · 3 marks

$k = \frac{1}{8}\ln(4/3)$.

e^{-8k}=3/4，ln后求k。

M1	正确比值
M1	自然对数
A1	k

(b) · 3 marks

Half-life $8\ln 2/\ln(4/3)$; quarter time $16\ln 2/\ln(4/3)$ days.

e^{-kt}=1/2或1/4，对数指数分别ln2与2ln2，所以时间加倍。

M1	半衰关系
M1	四分之一关系
A1	两时间

Question 9

P3-C05-R09 · 5 marks

(a) · 3 marks

$a=-3/2, B=e^{2}.$

lny=ax+lnB，直读斜率和对数截距，再指数还原B。

M1	正确线性形式
A1	a
A1	B

(b) · 2 marks

$x=4/3.$

lny=0给−3x/2+2=0。

M1	使用对数直线求零值
A1	x

Question 10

P3-C05-R10 · 6 marks

(a) · 4 marks

$x = \ln 4$; reject $e^x = -1/2$.

(2u+1)(u−4)=0，u=e^x严格正。

M1	指数代换分解
M1	排负u
M1	ln还原
A1	唯一根

(b) · 2 marks

$1/4$.

e^{-x}=1/e^x=1/4。

M1	指数倒数关系
A1	值

Question 11

P3-C05-R11 · 7 marks

(a) · 4 marks

Domain $\mathbb{R}$; range $[0, \infty)$; only intercept $(0, 0)$.

$x^2+1 \geqslant 1$，ln输出$\geqslant 0$，在x0取0，任意更大真数可取。

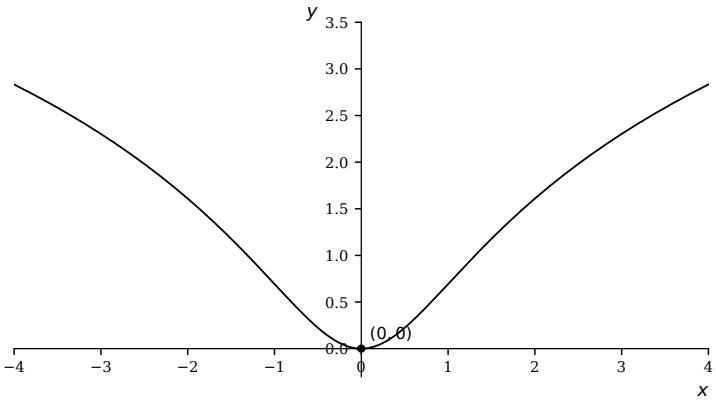
M1	真数恒正
M1	最小输出
A1	域与范围
B1	截距

(b) · 3 marks

Equality $x = \pm 2$; inequality $-2 < x < 2$.

ln单调，分别化为$x^2+1=5$或<5。

M1	合法去对数
M1	平方方程与不等式
A1	两个解集



Suggested sketch

Question 12

P3-C05-R12 · 7 marks

(a) · 2 marks

$k = \ln 2/3.$

e^{3k}=2，取ln。

M1	倍增条件
A1	k

(b) For this model, find

(i) · 2 marks

1600.

9年等于3个倍增周期，所以200×2^{3}。

M1	周期倍数
A1	数值

(ii) · 3 marks

7 years.

$t \geqslant 3\ln 5/\ln 2 \approx 6.966$，最小整数7，等号条件为至少。

M1	正确阈值不等式
M1	整数取上界
A1	年数

Question 13

P3-C05-R13 · 6 marks

(a) · 4 marks

$x = (1 + \sqrt{13})/2.$

$\ln(x+3)=\ln x^2$给$x^2-x-3=0$，正根唯一满足$x>0$。

M1	对数幂法则
M1	二次公式
M1	正域筛选
A1	根

(b) · 2 marks

$\ln x.$

由$x+3=x^2$，商为x，真数正。

M1	使用根关系化简商
A1	值

Question 14

P3-C05-R14 · 6 marks

(a) · 3 marks

$A = 4, B = -3$.

渐近线给B，e^{2ln2}=4，13=4A−3。

B1	B
M1	精确指数还原
A1	A

(b) · 3 marks

x-intercept $(\frac{1}{2}\ln(3/4), 0)$; y-intercept $(0, 1)$.

令4e^{2x}=3取ln；x0时4−3=1。

M1	指数零值方程
M1	对数求x
A1	两截距

Question 15

P3-C05-R15 · 6 marks

(a) · 3 marks

$(x+1)/(x-1) = 1 + 2/(x-1) > 1$, so its natural logarithm is positive.

利用x>1使分母正，真数严格大于1，ln单调且ln1=0。

M1	真数改写
M1	正值比较
A1	严格结论

(b) · 3 marks

$x = 3/2$.

(x+1)/(x−1)=5给x+1=5x−5，x1.5合法。

M1	去对数并清合法分母
M1	求解检查
A1	x

Question 16

P3-C05-R16 · 6 marks

(a) · 3 marks

$t = \frac{10}{3}\ln 4$.

相除4e^{-0.3t}=1，对数后t=ln4/.3。

M1	相比隔离指数
M1	ln运算
A1	精确时间

(b) · 3 marks

$0 \le t < \frac{10}{3}\ln 4$.

4e^{-0.3t}>1，随t严格减，所以在首次等值之前；包括初始时点。

M1	正确不等式方向
M1	原时间域
A1	区间

Question 17

P3-C05-R17 · 6 marks

(a) · 3 marks

$A = 5, k = \ln 2.$

首点A5，e^{2k}=4所以kln2。

B1	A
M1	对数求k
A1	k

(b) · 3 marks

$\ln y = \ln(5/2);\ y = 5/2.$

用lny=ln5+xln2，在−1处相减；也可代指数模型。

M1	正确对数线性值
M1	还原
A1	两值

Question 18

P3-C05-R18 · 6 marks

(a) · 4 marks

$0 < x < 2$; solution $x = 3/2$.

两真数均正；合并成ln[3(2−x)]，得到x=6−3x。

M1	两真数正域
M1	合并对数
M1	解并查域
A1	解

(b) · 2 marks

$\ln(3/4)$.

代1.5×.5=3/4。

M1	正确乘积代值
A1	对数值

Question 19

P3-C05-R19 · 7 marks

(a) · 4 marks

$R_0 = 10$, $k = \ln(3/2)/20$.

首条件读R0，第二e^{20k}=3/2后取ln。

B1	R0
M1	观测比例
M1	对数
A1	k

(b) · 3 marks

20 degrees; $R(T+\Delta)/R(T) = e^{k\Delta}$.

相除时初值和e^{kT}抵消，固定比值只由温差决定。

M1	正确比值式
M1	消去起始温度
A1	温差和理由

Question 20

P3-C05-R20 · 6 marks

(a) · 3 marks

$y = 2 + 6e^{-x/2}$.

ln(y−2)=−x/2+ln6，指数还原再加2。

M1	写线性式
M1	指数还原
A1	模型

(b) · 3 marks

$x = 2\ln 2$; asymptote $y = 2$.

6e^{-x/2}=3给e^{-x/2}=1/2，求x；指数衰减极限加回2。

M1	对数求x
A1	精确x
B1	原坐标渐近线

Question 21

P3-C05-R21 · 7 marks

(a) · 4 marks

$x = \pm \ln 2$.

设u=e^x>0，u+1/u=5/2，2u²−5u+2=0给u2或1/2。

M1	正变量代换
M1	二次因式
M1	ln还原
A1	两x

(b) · 3 marks

$\pm 3/2$.

可分别代u，或平方差关系：(sum)²−(difference)²=4，差平方9/4。

M1	代值或平方恒等式
M1	两符号
A1	值

Question 22

P3-C05-R22 · 7 marks

(a) · 4 marks

$k = \ln 4/4$; limit 100.

1−e^{-4k}=.75给e^{-4k}=.25；t大时指数趋0。

M1	校准方程
M1	对数
A1	k
B1	极限

(b) · 3 marks

$t = 8$.

此时指数=1/16，为t4处1/4的平方，所以时间8；或用ln求。

M1	正确指数比例
M1	求时间
A1	精确值

Question 23

P3-C05-R23 · 7 marks

(a) · 4 marks

$f^{-1}(x) = (e^x + 2)/3$, domain $\mathbb{R}$, range $(2/3, \infty)$.

原域x>2/3，原值域全实；反解e^y=3x−2，交换域范围。

M1	正确指数反解
A1	反函数
B1	逆域
B1	逆范围

(b) · 3 marks

$x = 1$.

f1=ln1=0，所以右边0，3x−2=1。

M1	先求f1
M1	反解零值
A1	x

Question 24

P3-C05-R24 · 7 marks

(a) · 4 marks

$x = e^{3}, e^{-2}$.

令u=lnx，(u−3)(u+2)=0；两实u都可指数还原为正x。

M1	对数代换
M1	因式分解
M1	指数回代
A1	两x

(b) · 3 marks

$e^{-2} \leq x \leq e^{3}$.

二次条件−2⩽lnx⩽3，指数单调，闭端点保留。

M1	二次范围
M1	单调指数化
A1	闭区间

Question 25

P3-C05-R25 · 6 marks

(a) · 3 marks

$(\frac{2}{3}\ln 3, 3^{4/3})$.

e^{3x}=9给x=ln9/3，纵坐标e^{2x}=9^{2/3}。

M1	合并指数
M1	ln求x
A1	精确交点

(b) · 3 marks

$\frac{2}{3}\ln 3$.

第二点x0，第一点横坐标正，差即该值。

M1	找零对数位置x0
M1	横差
A1	距离

Question 26

P3-C05-R26 · 6 marks

(a) · 4 marks

$A = 100, n = -3$; $y = 25/2$.

logy=logA+nlogx，截距2意味着A100而不是2；代x2。

M1	对数截距还原
B1	n
M1	代原模型
A1	y

(b) · 2 marks

Gradient -3; intercept $\ln 100 = 2\ln 10$.

lny=nlnx+lnA，换底不改变幂指数斜率，但截距依底而变。

M1	自然对数线性式
A1	两参数

Question 27

P3-C05-R27 · 6 marks

(a) · 3 marks

$x = \ln(e^x)$, so combine as a quotient $(e^x + 1)/e^x = 1 + e^{-x}$.

分母e^x严格正，日志商法则始终合法。

M1	将x改写为lne^x
M1	商化简
A1	目标式

(b) · 3 marks

$x = -\ln 2$.

1+e^{-x}=3给e^{-x}=2，求x。

M1	使用化简式
M1	指数解
A1	精确根

Question 28

P3-C05-R28 · 6 marks

(a) · 2 marks

3.79.

C5=8e^{-1.5}+2≈3.78504。

M1	正确代值
A1	精度

(b) · 4 marks

$t > \frac{10}{3}\ln 8 \approx 6.93$; boundary $(10/3)\ln 8$.

8e^{−.3t}<1，−.3t<−ln8，除负数方向翻转。

M1	正确严格指数条件
M1	不等号翻转
A1	精确边界
A1	数值

Question 29

P3-C05-R29 · 8 marks

(a) · 3 marks

$k = 0.3054$, $A = 110.5$.

两条件相除$e^{-3k} = 2/5$，所以$k = \frac{1}{3}\ln(5/2)$；再由$A = 60e^{2k} = 110.52\ldots$。

M1	比值消去A
A1	k
A1	A

(b) Using unrounded constants,

(i) · 2 marks

$t = 7.87$ hours.

用$t = 2 + \ln(60/10)/k = 2 + \ln 6/k = 7.866\ldots$，保持起算时刻为$t = 0$。

M1	对数时间方程
A1	绝对时刻

(ii) · 3 marks

$\ln 2/k = 2.27$ hours.

比值$M(t+T)/M(t) = e^{-kT} = 1/2$，所以$T = \ln 2/k$；$A$和起始时间都约去，因此与起始质量无关。

M1	比值及对数
A1	时间
B1	独立性理由

Question 30

P3-C05-R30 · 7 marks

(a) · 3 marks

$p = 3,\ q = 3/2.$

线性化为$\ln(y-2) = \ln p + q\ln x$。截距给$p = 3$；斜率$\ln(24/3)/\ln 4 = 3/2$，所以$y = 2 + 3x^{3/2}$。

M1	对数线性化
A1	p
A1	q

(b) For this model,

(i) · 2 marks

$x = 9.$

$2 + 3x^{3/2} = 83$给$x^{3/2} = 27$，所以$x = 9$。

M1	使用已求模型
A1	x

(ii) · 2 marks

$19.9\%.$

$y(1) = 5$，$y(1.21) = 2 + 3(1.21)^{3/2} = 5.993$。增幅$(5.993-5)/5 \times 100\% = 19.86\%$，即$19.9\%$。常数平移$2$使得$y$的百分比变化不同于$y-2$。

M1	输入倍率的幂
A1	百分比